

Empirical Proof Record: VALIDATED

2026-05-22T16:01:39.308346+00:00

Empirical Proof Record — VALIDATED

Proof ID: ef1cabdca834fb1cce611630f3e1091e...

Created: 2026-05-22T16:01:39.308346+00:00

1. Claim

Across a long single-pass stream, the substrate still recalls earlier experiences beyond a fixed context window: (for every probe whose lag exceeds `window_ref=256`, recall Jaccard ≥ 0.50). The reported floor is the worst beyond-window probe.

- **Operationalisation:** Stream 400 distinct stimuli once through Kimera's entity target, then re-probe 20 positions spread over the stream. For each probe, `recall = Jaccard(concept set at first exposure, concept set at probe)`; `lag = intervening cycles`. `recall_floor_beyond_window = min recall over probes with lag > 256`; the invariant holds iff that floor ≥ 0.50 .
- **Threshold:** `recall_floor_beyond_window >= 0.5 jaccard`
- **H0:** H0: `recall < 0.50` for some item recalled beyond `window_ref=256` — the horizon is shorter than the reference window
- **H1:** H1: `recall  $\geq 0.50$` for every beyond-window probe — memory reaches past a fixed window of that size

2. Verdict

VALIDATED — observed 1.0

recall floor 1.0000 over 5 beyond-window probes (lag > 256); mean recall 1.0000, stdev 0.0000; memory horizon (max lag with recall ≥ 0.50) = 360 cycles; max lag measured 360; 400-item single-pass stream; 5 probes produced no concept set; worst beyond-window probe: stimulus 42 @ lag 360 = 1.0000; control: beyond-window recall median 1.000 vs cross-item median 0.000, Mann-Whitney $p=7.5e-06$ (recall real beyond window)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.memory_horizon	recall_floor_beyond_window	1.0	—	ophamin 0.103.0

4. Pre-registration

- Registered at: 2026-05-22T16:01:39.048005+00:00
- Config hash: 44cc9a3bf33f3a0b7bdab2a0977ca76f...
- Data hash: ad79ab41ed3eb80cd1fbe8041eea8cc8...

5. Signature

f8a384a68785c069f9c6e4d2e11d03c54d73f00d8348a35fb27acc27e158719b